

Computational Microelectronics

Lecture 11

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L11

Homework#11

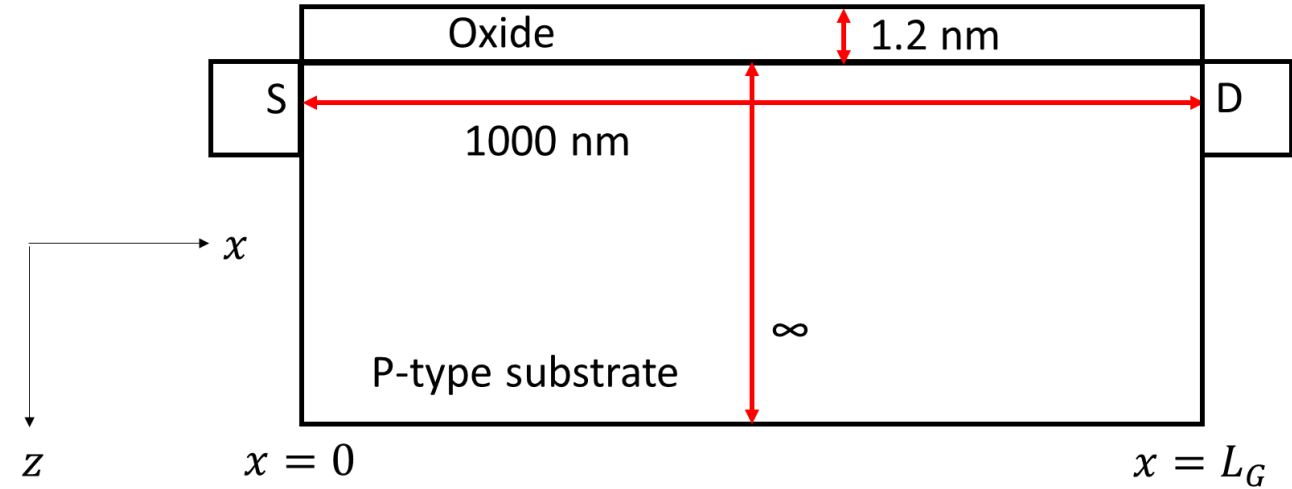
- Due: AM08:00, October 14 (One week later)
- Problem#1
 - Repeat Homework#7. This time, adopt a better implementation studied in L8 & L9.
 - Compare the convergence behavior of your two implementations (Homework#7 and Homework#11).
 - Test your codes with various N_x values.

Issues

- We have written a code to calculate I_D .
 - ~~1) Electron continuity equation is analytically solved.~~
 - ~~2) MOS equation and continuity equation are separately treated.~~
 - ~~3) Mobility is assumed to be constant. (Partially covered)~~
 - 4) Short-channel MOSFET
 - 5) Source/drain regions
 - 6) Transient effect
 - 7) Multiple devices (Can we calculate the VTC of a CMOS inverter?)

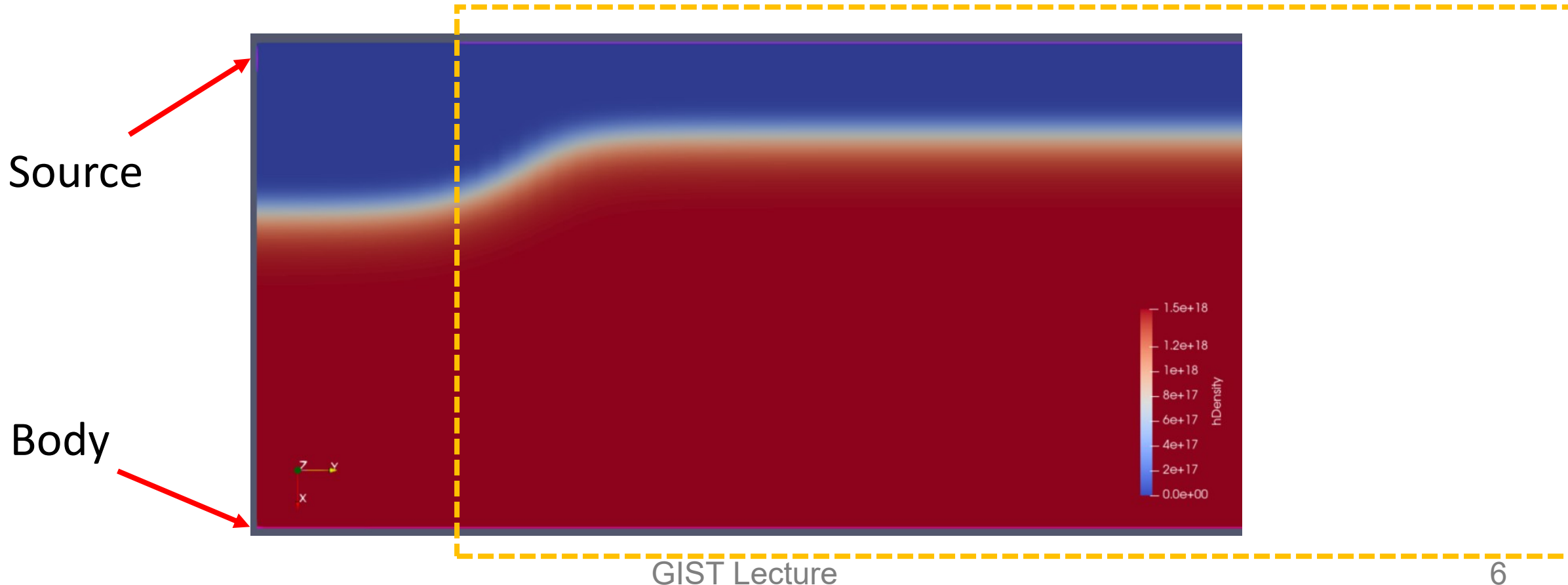
Source/drain regions are not included.

- From $x = 0$ to $x = L_G$
 - At $x = 0$, $\phi_n(0) = V_S$
 - At $x = L_G$, $\phi_n(L_G) = V_D$



Magnified view

- The real source contact is not located at $x = 0$.
 - What if there is a potential difference?



A better way

- Include the source/drain regions in the simulation domain.
 - The correct boundary condition is applied.
 - Source/drain series resistances are considered.



Poisson equation for source/drain

- We neglect the lateral field. (Gradual channel approx.)
 - When z is smaller than the junction depth X_j , we have the n-type doping density of N_D .
 - Beyond X_j , the p-type doping density of N_A .
 - Except for the position-dependent doping density, the problem is the same.

$$\frac{d^2\phi}{dz^2} = -\frac{q}{\epsilon_{si}} \left[n_{int} \exp\left(-\frac{q\phi}{k_B T}\right) - n_{int} \exp\left(\frac{q(\phi - \phi_n)}{k_B T}\right) + N_{dop}(z) \right]$$

- In the above equation, $N_{dop}(z) = +N_D$ for $z < X_j$ and $N_{dop}(z) = -N_A$ for $z > X_j$.

Change of variable

- Again, we introduce $\bar{\phi} = \phi - \phi_{\infty}$

$$\frac{d^2 \bar{\phi}}{dz^2} = -\frac{q}{\epsilon_{si}} \left[N_A \exp\left(-\frac{q\bar{\phi}}{k_B T}\right) - \frac{n_{int}^2}{N_A} \exp\left(-\frac{q\phi_n}{k_B T}\right) \exp\left(\frac{q\bar{\phi}}{k_B T}\right) + N_{dop}(z) \right]$$

– After multiplying $\frac{d\bar{\phi}}{dz}$,

$$\frac{1}{2} \frac{d}{dz} \left(\frac{d\bar{\phi}}{dz} \right)^2 = -\frac{q}{\epsilon_{si}} \left[N_A \exp\left(-\frac{q\bar{\phi}}{k_B T}\right) - \frac{n_{int}^2}{N_A} \exp\left(-\frac{q\phi_n}{k_B T}\right) \exp\left(\frac{q\bar{\phi}}{k_B T}\right) + N_{dop}(z) \right] \frac{d\bar{\phi}}{dz}$$

Result of integration from ∞ to z

- At any point z ,

$$\begin{aligned}
 & \frac{1}{2} \left(\frac{d\bar{\phi}}{dz} \right)^2 \\
 &= \frac{q}{\epsilon_{si}} \frac{k_B T}{q} \left[N_A \left\{ \exp \left(-\frac{q\bar{\phi}}{k_B T} \right) - 1 \right\} \right. \\
 & \quad \left. + \frac{n_{int}^2}{N_A} \exp \left(-\frac{q\phi_n}{k_B T} \right) \left\{ \exp \left(\frac{q\bar{\phi}}{k_B T} \right) - 1 \right\} \right] - \frac{q}{\epsilon_{si}} \int_{\infty}^z N_{dop}(z') \frac{d\bar{\phi}}{dz'} dz' \\
 &\approx \frac{q}{\epsilon_{si}} \frac{k_B T}{q} \left[-N_A + \frac{n_{int}^2}{N_A} \exp \left(-\frac{q\phi_n}{k_B T} \right) \left\{ \exp \left(\frac{q\bar{\phi}}{k_B T} \right) - 1 \right\} \right] \\
 & \quad - \frac{q}{\epsilon_{si}} \int_{\infty}^z N_{dop}(z') \frac{d\bar{\phi}}{dz'} dz'
 \end{aligned}$$

When $\bar{\phi} \gg \frac{k_B T}{q}$

Result of integration at $z = 0$

- Note that N_{dop} changes at $z = X_j$.

- Potential at $z = X_j$

$$\int_{-\infty}^0 N_{dop}(z') \frac{d\bar{\phi}}{dz'} dz' = - \int_0^{\bar{\phi}(X_j)} N_A d\bar{\phi} + \int_{\bar{\phi}(X_j)}^{\bar{\phi}(0)} N_D d\bar{\phi}$$
$$= -N_A \bar{\phi}(X_j) + N_D (\bar{\phi}(0) - \bar{\phi}(X_j)) = N_D \bar{\phi}(0) - (N_D + N_A) \bar{\phi}(X_j)$$

- Then, how can we find $\bar{\phi}(X_j)$?

- In general, we have to introduce another condition for $\bar{\phi}(X_j)$.

- We assume a neutral point where the electric field vanishes and the potential satisfies $N_D = \frac{n_{int}^2}{N_A} \exp\left(-\frac{q\phi_n}{k_B T}\right) \exp\left(\frac{q\bar{\phi}_{neutral}}{k_B T}\right)$.

Result of integration at the neutral point

- Doping integration

$$\begin{aligned} \int_{-\infty}^{Neutral} N_{dop}(z') \frac{d\bar{\phi}}{dz'} dz' &= - \int_0^{\bar{\phi}(X_j)} N_A d\bar{\phi} + \int_{\bar{\phi}(X_j)}^{\bar{\phi}_{neutral}} N_D d\bar{\phi} \\ &= -N_A \bar{\phi}(X_j) + N_D (\bar{\phi}_{neutral} - \bar{\phi}(X_j)) \\ &= N_D \bar{\phi}_{neutral} - (N_D + N_A) \bar{\phi}(X_j) \end{aligned}$$

– Then,

$$\begin{aligned} 0 &= \frac{q}{\epsilon_{si}} \frac{k_B T}{q} \left[-N_A + \frac{n_{int}^2}{N_A} \exp\left(-\frac{q\phi_n}{k_B T}\right) \left\{ \exp\left(\frac{q\bar{\phi}_{neutral}}{k_B T}\right) - 1 \right\} \right] \\ &\quad - \frac{q}{\epsilon_{si}} [N_D \bar{\phi}_{neutral} - (N_D + N_A) \bar{\phi}(X_j)] \end{aligned}$$

Difference between two integrations

- Surface electric field (Note that $\phi_s = \bar{\phi}(0)$.)

$$\begin{aligned}
 & \frac{1}{2} \left(\frac{d\phi}{dz} \Big|_0 \right)^2 \\
 &= \frac{q}{\epsilon_{si}} \frac{k_B T}{q} \left[\frac{n_{int}^2}{N_A} \exp \left(-\frac{q\phi_n}{k_B T} \right) \left\{ \exp \left(\frac{q\phi_s}{k_B T} \right) - \exp \left(\frac{q\bar{\phi}_{neutral}}{k_B T} \right) \right\} \right] \\
 & - \frac{q}{\epsilon_{si}} N_D (\phi_s - \bar{\phi}_{neutral}) \\
 &= \frac{q}{\epsilon_{si}} N_D \left[\frac{k_B T}{q} \left\{ \exp \left(\frac{q(\phi_s - \bar{\phi}_{neutral})}{k_B T} \right) - 1 \right\} - (\phi_s - \bar{\phi}_{neutral}) \right] \\
 & - \text{The sign of } \phi_s - \bar{\phi}_{neutral} \text{ determines the sign of the surface electric field.}
 \end{aligned}$$

Capacitance

- What is the surface electric field, $E_z(x > 0, z = 0)$?

– Then, the areal capacitance $C(x)$ becomes

$$C(x) = \epsilon_{ox} \frac{|E_z(x, 0)|}{V_G}$$

– Electric field? Calculate it from the electrostatic potential!

$z = -t_{ox}$ Gate = V_G

$z = 0$ Interface = 0 V $x = 0$

Thank you!